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RF Amplifier with Integrated Harmonic Termination Circuit

Abstract

A harmonic termination circuit is integrated into an RF amplifier semiconductor structure. A MIMcap is formed on a grounded source finger of an RF transistor amplifier, and one or more bond wires connect the MIMcap to a gate or drain bond pad of the transistor. The bond wire has an inherent inductance, and together with the MIMcap forms a series LC resonant circuit connected in shunt configuration from the gate or drain bond pad to ground. The LC circuit is tuned to shunt the desired harmonic component to ground, such as by adjusting the length of the bond wire (e.g., by altering its height). Such adjustment can be made after fabrication of the integrated circuit die is complete, and can be changed without requiring re-fabrication. The harmonic termination circuit resides entirely within the amplifier area, and does not increase the size of the die.

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Background/Summary

BACKGROUND

[0001] Electrical circuits requiring high power handling capability while operating at high frequencies, such as R-band (0.5-1 GHz), L-band (1-2 GHz), S-band (2-4 GHz), C-Band (4-8 GHz), X-band (8-12 GHz), Ku-band (12-18 GHz), K-band (18-27 GHz), Ka-band (27-40 GHz) and V-band (40-75 GHz) have become more prevalent. In particular, there is now a high demand for radio frequency (“RF”) transistor amplifiers that are used to amplify RF signals at frequencies of, for example, 500 MHz and higher (including microwave frequencies). These RF transistor amplifiers may need to exhibit high reliability, good linearity and handle high output power levels.

[0002] RF amplifiers are widely used in wireless communications systems and other applications. RF amplifiers are typically formed as semiconductor integrated circuit chips. Most RF amplifiers are implemented in silicon or using wide bandgap semiconductor materials, such as silicon carbide (“SiC”) and Group III nitride materials. As used herein, the term “Group III nitride” refers to those semiconducting compounds formed between nitrogen and the elements in Group III of the periodic table, usually aluminum (Al), gallium (Ga), and/or indium (In). The term also refers to ternary and quaternary compounds, such as GaN-based compounds AlGaN and AlInGaN. These compounds have empirical formulas in which one mole of nitrogen is combined with a total of one mole of the Group III elements.

[0003] Silicon-based RF amplifiers are typically implemented using laterally diffused metal oxide semiconductor (“LDMOS”) transistors. Silicon LDMOS RF amplifiers can exhibit high levels of linearity and may be relatively inexpensive to fabricate. Group III nitride-based RF amplifiers are typically implemented using High Electron Mobility Transistors (“HEMT”) and are primarily used in applications requiring high power and/or high frequency operation where LDMOS transistor amplifiers may have inherent performance limitations.

[0004] RF amplifiers may include one or more amplification stages, with each stage typically implemented as a transistor amplifier. In order to increase the output power and current handling capabilities, RF amplifiers are typically implemented in a “unit cell” configuration in which a large number of individual “unit cell” transistors are arranged electrically in parallel. The RF amplifier may be implemented as a single integrated circuit chip or “die,” or may include a plurality of dies. When multiple RF amplifier dies are used, they may be connected in series and/or in parallel.

[0005] RF amplifiers are typically designed to amplify RF signals at a fundamental frequency or range of frequencies. Ideally, for a transistor biased to class A operation, an output signal amplitude would be a constant multiple of the input amplitude:

V.sub.out=AV.sub.in

[0006] Due to inherent nonlinearities in transistor operation, the signal output by any real-world transistor amplifier includes not only the amplified RF signal at its fundamental frequency, but additionally signal components at multiples, or harmonics, of the fundamental frequency. A general equation for the actual output of a transistor biased to class A operation is:

[00001] $V_{out} = AV_{in} + BV_{in}^2 + CV_{in}^3 + $ .Math.

The second term of this equation is known as the second-order harmonic or distortion (and the third term is the third-order harmonic, etc.). Harmonic signal components cause constructive and destructive interference with the fundamental frequency signal, resulting in non-linearities in the amplifier's output which must be compensated (e.g., by pre-distortion). Harmonics also reduce the efficiency of RF amplifier operation.

[0007] Accordingly, harmonic termination circuits are well known in the art. Harmonic termination circuits are typically implemented as resonant circuits, which form a low-impedance path to RF signal ground at a targeted frequency or range of frequencies (e.g., 2.sup.nd or 3.sup.rd order harmonic frequencies). For example, U.S. Pat. No. 11,114,988 discloses a harmonic termination circuit integrated with the impedance inverter of a Doherty amplifier. The resonant circuits may be formed using transmission lines or discrete reactive components, such as inductors and capacitors. These components may be external ("off-chip"), which increases component count and limits RF amplifier deployment options to situations that can support off-chip components, such as a Printed Circuit Board (PBC).

[0008] Alternatively, the components can be formed on-chip. Inductors are formed in integrated circuits by routing metal traces in a spiral or serpentine pattern, close enough that the magnetic fields generated by current in the traces overlap. Capacitors can be formed on integrated circuits by depositing a metal layer as the bottom plate, depositing an insulating layer over the bottom plate, and depositing another metal layer over the insulating layer (known as a Metal-Insulator-Metal capacitor, or MIMcap). Forming these components increases the size, and therefore the cost, of the integrated circuit die. Additionally, the size and configuration of the reactive components must be precisely correct. That is, to alter the resonant frequency of a harmonic termination circuit, the inductor and/or capacitor structure must be redesigned, and the die re-fabricated, further increasing the cost.

[0009] The Background section of this document is provided to place embodiments of the present invention in technological and operational context, to assist those of skill in the art in understanding their scope and utility. Unless explicitly identified as such, no statement herein is admitted to be prior art merely by its inclusion in the Background section

BRIEF SUMMARY

[0010] The following presents a simplified summary of the disclosure in order to provide a basic understanding to those of skill in the art. This summary is not an extensive overview of the disclosure and is not intended to identify key/critical elements of embodiments of the invention or to delineate the scope of the invention. The sole purpose of this summary is to present some concepts disclosed herein in a simplified form as a prelude to the more detailed description that is presented later.

[0011] According to aspects of the present disclosure described and claimed herein, a harmonic termination circuit comprises a MIMcap integrated into a source finger of an RF transistor amplifier, and one or more bond wires connecting the MIMcap to a gate or drain bond pad of the transistor. The source terminal is electrically connected to RF signal ground, such as through a plated via to a metal layer on the back side of the die substrate. The bond wire has an inherent inductance, and together with the MIMcap forms a series LC resonant circuit connected in shunt configuration from the gate or drain bond pad to ground. The LC circuit is tuned to shunt the desired harmonic component to ground, such as by adjusting the length of the bond wire (e.g., by altering its height). Note that such adjustment can be made after fabrication of the integrated circuit die is complete, and can be changed without requiring re-fabrication. Because the MIMcap is formed on the source finger, and connected to a gate or drain bond pad, the harmonic termination

circuit resides entirely within the amplifier area, and does not increase the size of the die.
[0012] One aspect relates to a semiconductor transistor configured to amplify a Radio Frequency (RF) signal at a fundamental frequency. The transistor includes a substrate having upper and lower surfaces, and a unit cell semiconductor transistor structure formed on the upper surface of the substrate. The transistor also includes source and drain fingers formed on, and electrically connected to, the semiconductor transistor structure. The transistor further includes gate fingers formed over the semiconductor transistor structure in active areas between the source and drain fingers. The transistor additionally includes gate and drain bond pads connected to respective gate and drain fingers. The source fingers are electrically connected to RF signal ground. A series resonant circuit is connected between one of the gate and drain bond pads and a source finger. The series resonant circuit comprises a Metal-Insulator-Metal (MIM) capacitor formed on the source finger, and an inductance implemented as a bond wire connected between the gate or drain bond pad and the MIM capacitor.
[0013] Another aspect relates to a method of manufacturing a semiconductor transistor configured to amplify a Radio Frequency (RF) signal at a fundamental frequency. A substrate having upper and lower surfaces is provided. A unit cell semiconductor transistor structure is formed on the upper surface of the substrate. Source and drain fingers are formed on, and electrically connected to, the semiconductor transistor structure. Gate fingers are formed over the semiconductor transistor structure in active areas between the source and drain fingers. Gate and drain bond pads, connected to respective gate and drain fingers, are formed. The source fingers are electrically connected to RF signal ground. A series resonant circuit, connected between one of the gate and drain bond pads and a source finger, is formed by forming a Metal-Insulator-Metal (MIM) capacitor on the source finger, and connecting a bond wire between the gate or drain bond pad and the MIM capacitor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] FIG. **1** is a top section view of an RF transistor amplifier device.
[0015] FIG. **2** is a side section view of the RF transistor amplifier device.
[0016] FIG. **3** is a schematic electrical diagram of an amplifier with a harmonic termination circuit.
[0017] FIG. **4**A is top sectional view of harmonic termination circuits connected to a gate pad.
[0018] FIG. **4**B is top sectional view of some harmonic termination circuits connected to a gate pad and others to a drain pad.
[0019] FIG. **4**C is top sectional view of harmonic termination circuits connected to a drain pad, with an alternate location of MIMcaps on the source fingers.
[0020] FIG. **4**D is top sectional view of harmonic termination circuits connecting each source finger to both a gate pad and a drain pad.
[0021] FIG. **5** is a flow diagram of a method of manufacturing an RF amplifier.
[0022] FIG. **6** is a side section view of the RF transistor amplifier device.
[0023] FIG. **7** is a side section view of the RF transistor amplifier device in a first packaging option.
[0024] FIG. **8** is a side section view of the RF transistor amplifier device in a second packaging option.
[0025] FIG. **9** is a side section view of the RF transistor amplifier device in a third packaging option.
DETAILED DESCRIPTION
[0026] The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing

figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.

[0027] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0028] It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

[0029] Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.

[0030] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0031] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0032] Reference now will be made in detail to aspects of the present disclosure, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the aspect, not limitation of the present disclosure. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made to the aspects without departing from the scope or spirit of the present disclosure. For instance, features illustrated or described as part of one aspect can be used with another aspect to yield a still further aspect. Thus, it is intended that aspects of the present disclosure cover such modifications and variations.

[0033] Transistor devices, such as High Electron Mobility Transistors (HEMT), may be used in power electronics applications. HEMTs fabricated in Group III nitride-based material systems may have the potential to generate large amounts of radio frequency (RF) power because of the combination of material characteristics that includes high breakdown fields, wide bandgaps, large conduction band offset, and/or high saturated electron drift velocity. As such, Group III nitride based HEMTs may be promising candidates for high frequency and/or high-power RF applications, as well as for low frequency high power switching applications, both as discrete transistors or as coupled with other circuit elements, such as in Monolithic Microwave Integrated Circuit (MMIC) devices.

[0034] Transistor devices such as HEMT devices may be classified into depletion mode and enhancement mode types, corresponding to whether the transistor is in an ON-state or an OFF-state at a gate-source voltage of zero. In enhancement mode devices, the devices are OFF at zero gate-source voltage, whereas in depletion mode devices, the device is ON at zero gate-source voltage. Often, high performance Group III nitride-based HEMT devices may be implemented as depletion mode (normally-on) devices, in that they are conductive at a gate-source bias of zero due to the polarization-induced charge at the interface of the barrier and channel layers of the device.

[0035] When an HEMT device is in an ON-state, a 2-Dimensional Electron Gas (2DEG) is formed at the heterojunction of two semiconductor materials with different bandgap energies, where the smaller bandgap material has a higher electron affinity. The 2DEG is an accumulation layer in the smaller bandgap material and can include a very high sheet electron concentration. Additionally, electrons that originate in the wider-bandgap semiconductor material transfer to the 2DEG layer, allowing high electron mobility due to reduced ionized impurity scattering. This combination of high carrier concentration and high carrier mobility may give the HEMT device a very large transconductance (which may refer to the relationship between output current and input voltage) and may provide a strong performance advantage over MOSFETs for high-frequency applications.

[0036] Aspects of the present disclosure are described herein with reference to plan view and cross-section illustrations that are schematic illustrations of idealized embodiments (and intermediate structures) of aspects of the disclosure. The thickness of layers and regions in the drawings may be exaggerated for clarity. Additionally, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, aspects of the disclosure should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing. Similarly, it will be understood that variations in the dimensions are to be expected based on standard deviations in manufacturing procedures. As used herein, “approximately” or “about” includes values within 10% of the nominal value.

[0037] Like numbers refer to like elements throughout. Thus, the same or similar numbers may be described with reference to other drawings even if they are neither mentioned nor described in the corresponding drawing. Also, elements that are not denoted by reference numbers may be described with reference to other drawings.

[0038] Some aspects of the disclosure are described with reference to semiconductor layers and/or regions which are characterized as having a conductivity type such as n type or p type, which refers to the majority carrier concentration in the layer and/or region. Thus, N type material has a majority equilibrium concentration of negatively charged electrons, while P type material has a majority equilibrium concentration of positively charged holes.

[0039] Aspects of the present disclosure are discussed with reference to an HEMT transistor device for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will appreciate that certain aspects of the present disclosure may be applicable to other transistor devices without deviating from the scope of the present disclosure. For instance, aspects of the present disclosure may be implemented in any transistor having a field plate or other transistors devices with metal structures, such as silicon carbide-based MOSFETs.

[0040] In the drawings and specification, there have been disclosed typical embodiments and, although specific terms are employed, they are used in a generic and descriptive sense only and not for purposes of limitation of the scope set forth in the following claims.

[0041] With reference now to the Figures, example aspects of the present disclosure will now be set forth. These aspects are described herein with reference to a Gallium Nitride (GaN) High Electron Mobility Transistor (HEMT), but are not limited to such implementation. For example, aspects of the present disclosure may advantageously be implemented using LDMOS or Gallium Arsenide (GaAs). To provide a full and complete disclosure enabling one of skill in the art to make and use the present disclosure, a description of a GaN HEMT is first presented. The discussion will then shift to the metal layers overlying a semiconductor transistor structure (again, using GaN HEMT as merely an example), which provide conductivity for interconnection with other circuitry.
[0042] FIG. **1** depicts a top section view of an RF power amplifier, which for the purposes of explanation herein comprises a Group III nitride-based HEMT RF amplifier **100**. The HEMT device **100** has a plurality of unit cell transistors **102** that each include a gate finger **104**, a drain finger **106**, and a source finger **108**. The gate fingers **104** are electrically connected to a common gate pad **110**, and the drain fingers **106** are electrically connected to a common drain pad **112**. The source fingers **108** are electrically connected to a device source terminal via a plurality of conductive source vias **114** that extend through the semiconductor layer structure **116**. The conductive source vias **114** may comprise metal-plated vias that extend completely through the semiconductor substrate **116**. The conductive source vias **114** may electrically connect to a metal layer covering the back side of the semiconductor substrate **116**, which may be connected to RF signal ground. At least the gate pad **110**, drain pad **112**, and source fingers **108** are the highest, or uppermost, layers of the semiconductor device **100**.
[0043] According to aspects of the present disclosure, a harmonic termination circuit **118** is formed at the output of at least one, and up to all, of the unit cell transistors **102**. The harmonic termination circuit **118** comprises a MIMcap **122** formed on the source finger **108** and a wire bond **120** connecting the MIMcap **122** to, in the case depicted, the drain pad **112**.
[0044] FIG. **2** depicts a side section view of the RF amplifier **100**, as indicated by the section line in FIG. **1**. FIG. **2** is intended to represent structures for identification and description and is not intended to represent the structures to physical scale. FIG. **2** shows the RF amplifier **100** comprising a substrate **116**, with a semiconductor transistor structure **12** formed over the substrate **116** in an active area, as explained in greater detail herein. The semiconductor transistor structure **12** may comprise a Group III-nitride HEMT structure.
[0045] A back side metal plate **30** is formed on the opposite side of the substrate **116** from the semiconductor transistor structure **12**. The back side metal plate **30** may electrically connect to the plated vias **114**, which extend through the substrate **116**. The back side metal plate **30** connects to a device source terminal, which may be electrically connected to RF signal ground. The gate pad **110** and drain pad **112** are formed on the substrate **116** (or on intervening layers). The source finger **108** is formed on the substrate **116**, over the active area of the semiconductor transistor structure **12**. The source finger **108** covers, and is electrically connected to, one or more plated vias **114**.
[0046] As described above, the output harmonic termination circuit **118** comprises a MIMcap **122** formed on the source finger **108**, connected to the drain pad **112** by one or more bond wires **120**. The MIMcap **122** is formed by depositing a dielectric layer **124** directly on at least a portion of the source finger **108**, and depositing a metal layer **126** directly on the dielectric layer **124**. The source finger **108** forms the bottom plate of the MIMcap **122**, with the metal layer **126** forming the top plate. The source finger **108**, and hence the MIMcap **122**, is connected to RF signal ground by the plated via(s) **114**. The MIMcap **122** is thus a capacitor connected to ground. As known in the art, the capacitance of the MIMcap **122** may be adjusted by the size of the metal layer **126**, the thickness of the dielectric layer **124**, the dielectric constant, and the like.
[0047] The bond wire(s) **120** connecting the MIMcap **122** to the drain pad **112** have an inherent parasitic inductance. The inductance of the bond wire(s) **120** may be adjusted by adjusting the length, which may be adjusted by varying the points along the drain pad **112** and MIMcap **122** at which the wire is bonded. The inductance may further be adjusted by altering the height of the

“loop” the bond wire **120** forms, as indicated by dotted lines in FIG. **2**, which also varies its length.

[0048] FIG. **3** is an electrical schematic representation of the RF amplifier **100** of FIGS. **1** and **2**. The amplifier **100** comprises a transistor having gate, drain, and source terminals, wherein the source terminal is electrically connected to RF signal ground. An output harmonic termination circuit **118** is coupled between the drain terminal and ground. The harmonic termination circuit **118** comprises a series LC resonant circuit comprising an inductance **120** and capacitance **122**. The series LC circuit is configured to present a high impedance at a fundamental operating frequency of the amplifier **100**, and a low impedance at a selected harmonic of the fundamental frequency. Hence, the harmonic termination circuit **118** has no effect on signals at the fundamental frequency, but shunts signals at the harmonic frequency to ground.

[0049] Those of skill in the art will readily recognize that a harmonic termination circuit **118** may alternatively or additionally be configured as an input harmonic termination circuit, wherein the bond wire(s) **120** connects the gate pad **110** to the MIMcap **122**. Both input and output harmonic termination circuits **118** may be provided, in any given implementation. Input and output harmonic termination circuits **118** may be combined in any combination.

[0050] As exemplary and non-limiting examples, FIGS. **4**A-D depict some of these variations. FIG. **4**A depicts input harmonic termination circuits **118**, wherein MIMcaps **122** formed on the source fingers **108** are connected by bond wires **120** to the gate bond pad **110**. Although three input harmonic termination circuits **118** are depicted, any number may be provided, as required or desired. Where more than one input harmonic termination circuit **118** is provided, they may be tuned to different harmonics of the fundamental frequency (i.e., the 2.sup.nd harmonic, 3.sup.rd harmonic, etc.).

[0051] FIG. **4**B shows both input and output harmonic termination circuits **118**, wherein one MIMcap **122** is formed on the source fingers **108**, and each is connected by bond wires **120** to either the gate bond pad **110** and source bond pad **112**. The harmonic termination circuits **118** may be tuned to the same, or different, harmonics of the fundamental frequency.

[0052] FIG. **4**C shows output harmonic termination circuits **118**, wherein MIMcaps **122** formed on the source fingers **108** are connected by bond wires **120** to the drain bond pad **110**. Note that in this example, the MIMcaps **122** are formed on a more distant region of the source fingers **108**—that is, spaced apart from the drain bond pad **112**. This may allow for a longer bond wire **120**, increasing the inductance in the series resonant circuit that forms each harmonic termination circuit **118**.

[0053] Finally, FIG. **4**D shows both input and output harmonic termination circuits **118** connected to each source finger **108**. In this case, two separate and electrically isolated (other than both being grounded) MIMcaps **122** are formed on each source finger **108**. One such MIMcap **122** is connected by a bond wire **120** to the gate bond pad **110**, and the other MIMcap **122** is connected by a bond wire **120** to the source bond pad **112**.

[0054] Those of skill in the art will readily recognize that the configurations shown in FIGS. **4**A-D are representative only, and are not limiting. In general, a broad range of configurations of RF transistors **100** incorporating harmonic termination circuits **118** are possible within the scope of the present disclosure.

[0055] FIG. **5** depicts the steps in a method **200** of manufacturing a semiconductor transistor **100** configured to amplify a Radio Frequency (RF) signal at a fundamental frequency. A substrate **116** having upper and lower surfaces is provided (block **210**). A unit cell semiconductor transistor structure **12** is formed on the upper surface of the substrate **116** (block **220**). Source and drain fingers **106**, **108** are formed directly on, and electrically connected to, the semiconductor transistor structure **12** (block **230**). Gate fingers **104** are formed over the semiconductor transistor structure **12** in active areas between the source and drain fingers **106**, **108** (block **240**). Gate and drain bond pads **110**, **112**, connected to respective gate and drain fingers **104**, **106**, are formed (block **250**). The source fingers **108** are electrically connected to RF signal ground (block **260**). A series resonant circuit **118**, connected between one of the gate and drain bond pads **110**, **112** and a source finger

108, is formed by forming a Metal-Insulator-Metal (MIM) capacitor **122** on the source finger **108**, and connecting a bond wire **120** between the gate or drain bond pad **110**, **112** and the MIM capacitor **122** (block **270**).

[0056] FIG. **6** depicts a side section view of a unit cell transistor **102** of the HEMT device **100**, as indicated by the section line in FIG. **1**. FIG. **6** is intended to represent structures for identification and description and is not intended to represent the structures to physical scale. The unit cell transistor **102** of the HEMT device **100** includes a semiconductor transistor structure **12**. The semiconductor transistor structure **12**, in this representative example, is a Group III-nitride HEMT structure.

[0057] As used herein, the term "Group III nitride" refers to those semiconducting compounds formed between nitrogen (N) and the elements in Group III of the periodic table, usually aluminum (Al), gallium (Ga), and/or indium (In). The term also refers to ternary and quaternary (or higher) compounds such as, for example, AlGaN and AlInGaN. As is well understood by those in this art, the Group III elements may combine with nitrogen to form binary (e.g., GaN), ternary (e.g., AlGaN, AlInN), and quaternary (e.g., AlInGaN) compounds. These compounds all have empirical formulas in which one mole of nitrogen is combined with a total of one mole of the Group III elements.

[0058] The semiconductor transistor structure **12** is formed on a substrate **116**. The substrate **116** may be a semiconductor material. For instance, the substrate **116** may be a silicon substrate, a silicon carbide (SiC) substrate, a sapphire substrate, or other suitable substrate. In some aspects, the substrate **116** may be a semi-insulating SiC substrate that may be, for example, the 4H polytype of silicon carbide. Other SiC candidate polytypes may include the 3C, 6H, and 15R polytypes. The substrate may be a High Purity Semi-Insulating (HPSI) substrate, available from Wolfspeed, Inc. The term "semi-insulating" is used descriptively herein, rather than in an absolute sense.

[0059] In some aspects of the present disclosure, the SiC bulk crystal of the substrate **116** may have a resistivity equal to or higher than about 1×10.sup.5 ohm-cm at room temperature. Example SiC substrates that may be used in some embodiments are manufactured by, for example, Wolfspeed, Inc., and methods for producing such substrates are described, for example, in U.S. Pat. No. Re. 34,861, U.S. Pat. Nos. 4,946,547; 5,200,022; and 6,218,680; the disclosures of which are incorporated by reference herein in their entireties. Although SiC may be used as a substrate material, aspects of the present disclosure may utilize any suitable substrate, such as sapphire (Al2O3), aluminum nitride (AlN), aluminum gallium nitride (AlGaN), gallium nitride (GaN), silicon (Si), GaAs, LGO, zinc oxide (ZnO), LAO, indium phosphide (InP), and the like. The substrate **116** may be a SiC wafer, and the HEMT device **100** may be formed, at least in part, via wafer-level processing, and the wafer may then be diced to provide a plurality of individual HEMT devices **100** that may include one or more transistor cells **102**.

[0060] In some aspects of the present disclosure, the substrate **116** of the HEMT device **100** may be a thinned substrate **116**. In some embodiments, the thickness of the substrate **116** may be about 100 μm or less, such as about 75 μm or less, such as about 50 μm or less.

[0061] The semiconductor transistor structure **12** includes a channel layer **16** on an upper surface **116**A of the substrate **116** (or on optional layers, not shown in FIG. **1**, deposited over the substrate, such as an optional buffer or nucleation layer). The semiconductor transistor structure **12** also includes a barrier layer **18** on an upper surface of the channel layer **16**. In some aspects, the channel layer **16** and the barrier layer **18** may each be formed by epitaxial growth. Techniques for epitaxial growth of Group III nitrides have been described in, for example, U.S. Pat. Nos. 5,210,051, 5,393,993, and 5,523,589, the disclosures of which are incorporated by reference herein in their enteritis. The channel layer **16** has a bandgap that is less than the bandgap of the barrier layer **18**. The channel layer **16** has a larger electron affinity than the barrier layer **18**. The channel layer **16** and the barrier layer **18** include Group III nitride-based materials.

[0062] In some aspects, the channel layer **16** is a Group III nitride, such as Al.sub.xGa.sub.1-xN, where x is about 0.1 or less, provided that the energy of the conduction band edge of the channel layer **16** is less than the energy of the conduction band edge of the barrier layer **18** at the interface between the channel layer **16** and barrier layer **18** (also known as a heterojunction). In some embodiments, the aluminum mole fraction x is approximately 0 (e.g., less than 5%, such as 0%), indicating that the channel layer **16** is GaN. The channel layer **16** may or may not include other Group III-nitrides such as InGaN, AlInGaN or the like. The channel layer **16** may be undoped ("unintentionally doped"). In some aspects, the channel layer **16** may be doped, for instance with iron (Fe). The channel layer **16** may be a multi-layer structure, such as a superlattice or combinations of GaN, AlGaN, or the like. The channel layer **16** may be under compressive strain in some aspects. In some aspects, the channel layer **16** can include an implanted region extending into the channel layer **16**. The implanted region may have implanted dopants to provide, for instance, N-type doping of the semiconductor structure in the implanted region, such as in the region on which source and drain terminals are formed.

[0063] The barrier layer **18** includes a Group III nitride-based layer having a surface **18**A positioned on a surface **16**A of the channel layer **16**. The barrier layer **18** is a Group III-nitride, such as Al.sub.yGa.sub.1-yN, where y is the aluminum mole fraction in the barrier layer **18**. In some embodiments, the aluminum mole fraction y is such that y is in a range of about 0.25 to about 0.40 (e.g., the aluminum mole fraction is in a range of about 25% to about 40%), indicating that the barrier layer is an AlGaN layer. For instance, in some embodiments, the aluminum mole fraction y is such that y is in a range from about 0.25 to about 0.35, such as about 0.25 to 0.30, such as about 0.30 (e.g., the aluminum mole fraction is in a range of about 25% to about 35%, such as about 25% to about 30%, such as about 30%). Additionally or alternatively, in some aspects, the aluminum mole fraction y is such that y is in a range of about 0.30 to about 0.40, such as about 0.35 to about 0.40, such as about 0.35 (e.g., the aluminum mole fraction is in a range of 30% to 40%, such as in a range of about 35% to about 40%, such as about 35%). For instance, in some implementations, the barrier layer **18** can include a higher aluminum mole fraction to achieve low contact resistance for high frequency applications.

[0064] The energy of the conduction band edge of the barrier layer **18** is greater than the energy of the conduction band edge of the channel layer **16** at the interface between the channel layer **16** and barrier layer **18**. The barrier layer **18** may include other Group III elements (e.g., In) without deviating from the scope of the present disclosure. The barrier layer **18**, in some examples, may be a multilayer structure. The multilayer structure may include multiple Group III nitride-based layers with differing aluminum mole fractions.

[0065] The barrier layer **18** may have a thickness less than about 130 Angstroms, such as in a range of about 10 Angstroms to about 130 Angstroms. For instance, in some aspects, the barrier layer can have a thickness in a range of about 70 Angstroms to about 100 Angstroms, such as about 80 Angstroms to about 90 Angstroms. For instance, in some implementations, the barrier layer **18** can have a reduced thickness to achieve low contact resistance for high frequency applications. The channel layer **16** and/or the barrier layer **18** may be deposited, for example, by metal-organic chemical vapor deposition (MOCVD), molecular beam epitaxy (MBE), or hydride vapor phase epitaxy (HVPE).

[0066] A 2DEG **20** is induced in the channel layer **16** at an interface between the channel layer **16** and the barrier layer **18**. The 2DEG **20** is highly conductive and allows conduction between the source and drain regions of the HEMT device **100**.

[0067] While the HEMT unit cell transistor **102** of FIG. **4** is shown with a substrate **116**, channel layer **16** and barrier layer **18** for purposes of illustration, the HEMT device **100** may include additional layers/structures/elements. For instance, the HEMT device **100** may include a buffer layer(s)/nucleation layer(s)/transition layer(s) (not shown) between substrate **116** and the channel layer **16**. For example, an AlN buffer layer may be on the upper surface **116**A of the substrate **116** to provide an appropriate crystal structure transition between a SiC substrate **116** and the channel

layer **16**. The optional buffer/nucleation/transition layers may be deposited by MOCVD, MBE, and/or HYPE.
[0068] The HEMT device **100** may include a cap layer (not shown) on the barrier layer **18**. HEMT structures including substrates, channel layers, barrier layers, and other layers are discussed by way of example in U.S. Pat. Nos. 5,192,987; 5,296,395; 6,316,793; 6,548,333; 7,544,963; 7,548,112; 7,592,211; 7,615,774; 7,709,269; 7,709,859; and 10,971,612, the disclosures of which are incorporated by reference herein in their entireties. Additionally, strain balancing transition layer(s) may also and/or alternatively be provided as described, for example, in U.S. Pat. No. 7,030,428, the disclosure of which is incorporated by reference herein.
[0069] The unit cell transistor **102** of the HEMT device **100** includes a source finger **108** on the semiconductor transistor structure **12**. The unit cell transistor **102** of the HEMT device **100** also includes a drain finger **106** on the semiconductor transistor structure **12**. The source finger **108** and the drain finger **106** are laterally spaced apart from each other. The source finger **108** and the drain finger **106** comprise an ohmic metal, which forms an ohmic contact to a Group III nitride-based semiconductor material. Suitable metals include refractory metals, such as titanium (Ti), tungsten (W), titanium tungsten (TiW), silicon (Si), titanium tungsten nitride (TiWN), tungsten silicide (WSi), rhenium (Re), niobium (Nb), nickel (Ni), gold (Au), aluminum (Al), tantalum (Ta), molybdenum (Mo), NiSi.sub.x, titanium silicide (TiSi), titanium nitride (TiN), tungsten silicon nitride (WSiN), platinum (Pt) and the like. In some aspects, the source finger **108** and/or the drain finger **106** may include a plurality of layers to form an ohmic contact as described, for example, in U.S. Pat. Nos. 8,563,372 and 9,214,352, the disclosures of which are incorporated by reference herein in their entireties.
[0070] The unit cell transistor **102** of the HEMT device **100** includes a gate finger **104** on the semiconductor transistor structure **12**. The gate finger **104** has a gate length L.sub.G. The gate length L.sub.G is the length of the gate contact **104** along the portion of the gate contact **104** that is on the semiconductor transistor structure **12** (e.g., the length of the lowermost portion of the gate finger **104** in contact with the semiconductor transistor structure **12**). In some aspects, the gate length L.sub.G may be about 200 nm or less, such as about 100 nm or less, such as in a range of about 60 nm to about 100 nm, such as in a range of about 70 nm to about 90 nm. A distance Lgd between the gate finger **104** and the drain finger **106** may be, for instance, in a range of 1.8 μm to about 2.2 μm, such as about 1.98 μm. A distance Lgs between the gate finger **104** and the source finger **108** may be, for instance, in a range of about 0.4 μm to about 0.8 μm, such as about 0.6 μm.
[0071] The material of the gate finger **104** may be chosen based on the composition of the barrier layer **18**, and may, in some aspects, be a Schottky contact. Materials capable of making a Schottky contact to a Group III nitride-based semiconductor material may be used, such as, for example, nickel (Ni), platinum (Pt), nickel silicide (NiSi.sub.x), copper (Cu), palladium (Pd), chromium (Cr), tungsten (W) and/or tungsten silicon nitride (WSiN).
[0072] In some aspects of the present disclosure, such as that depicted in FIG. **6**, the gate finger **104** is a T-shaped gate and/or a gamma gate, the formation of which is discussed by way of example in U.S. Pat. Nos. 8,049,252; 7,045,404; and 8,120,064, the disclosures of which are incorporated by reference herein in their entireties. The gate finger **104** may have an overhang toward the drain finger **106**. The length Γ.sub.D of the overhang toward the drain finger **106** may be in a range of about 0.15 μm to about 0.25 μm, such as about 0.2 μm. The gate finger **104** may have an overhang toward the source finger **108**. The length Γ.sub.S of the overhang toward the source finger **108** may be in a range of about 0.15 μm to about 0.25 μm, such as about 0.2 μm.
[0073] As FIG. **6** shows, the source finger **108** is formed over, and electrically connected to, a plated via **114** that extends from a lower surface **116**B of the substrate **116**, through the substrate **116** and semiconductor transistor structure **12** to the upper surface of the semiconductor transistor structure **12**. A back metal layer **30** is formed on the lower surface **116**B of the substrate **116** and on side walls of the via **114**. The back metal layer **30** is conductively coupled to the source finger **108**.

The back metal layer **30** may couple to RF signal ground.
[0074] In some aspects of the present disclosure, the via **114** has an oval or circular cross-section when viewed in a plan view (e.g., FIG. **1**). However, the present disclosure is not limited thereto. In some aspects of the present disclosure, a cross-section of the via **114** may be a polygon or other shape, as will be understood by one of ordinary skill in the art using the disclosures provided herein. In some aspects of the present disclosure, dimensions of the via **114** (e.g., a length and/or a width) may be such that the largest cross-sectional area of the via **114** is about 1000 μm.sup.2 or less. The cross-sectional area may be taken in a direction that is parallel to the lower surface **14**B of the substrate **116** (e.g., the X-Y plane of FIG. **1**). In some aspects of the present disclosure, the largest cross-sectional area of the via **114** may be that portion of the via **114** that is adjacent the lower surface **116**B of the substrate **116** (e.g., the opening of the via **114**). For example, in some aspects of the present disclosure, a greatest width (e.g., in the X direction in FIG. **6**) may be about 16 μm and a greatest length (e.g., in the Y direction in FIG. **6**) may be about 40 μm, though the present disclosure is not limited thereto. In some aspects of the present disclosure, sidewalls of the via **114** may be inclined and/or slanted with respect to the lower surface **116**B of the substrate **116**. In some aspects of the present disclosure, the sidewalls of the via **114** may be approximately perpendicular to the lower surface **116**B of the substrate **116**.
[0075] As FIG. **6** shows, a dielectric layer **124** is deposited over at least part of the source finger **108**. A metal layer **126** is deposited over the dielectric layer **124**. The source finger **108**, dielectric layer **124**, and metal layer **126** form a MIMcap **122**, which is electrically connected to RF signal ground via the source finger's **108** connection to the via(s) **114**. A bond wire **120** is attached to the upper surface of the metal layer **126** of the MIMcap **122**. The bond wire **120** connects to one of a gate bond pad **110** and a drain bond pad **112**. The gate and drain bond pads **110**, **112** are not shown in the section view of FIG. **6**, as they are out of the plane of the drawing figure (i.e., in front of or behind it). The MIMcap **122** and bond wire **120** form a harmonic termination circuit **118** comprising a shunt series resonant LC circuit connected between either the gate bond pad **110** or the drain bond pad **112** and RF signal ground. The series resonant LC circuit may be tuned to present a high impedance at a fundamental frequency of the RF amplifier **100**, and a low impedance at a selected harmonic of the fundamental frequency.
[0076] The unit cell transistor **102** of the HEMT device **100** includes a first dielectric layer **32** on the semiconductor transistor structure **12**. The first dielectric layer **32** directly contacts the upper surface of the semiconductor transistor structure **12**. At least a portion of the first dielectric layer **32** is between the semiconductor transistor structure **12** and at least a portion of the gate finger **104**. For instance, at least a portion of the first dielectric layer **32** may be between the semiconductor transistor structure **12** and the overhang of the gate finger **104**. In some aspects of the present disclosure, the thickness of the first dielectric layer **32** may be about 1450 Angstroms or less, such as in a range of about 800 Angstroms to about 1450 Angstroms, such as about 1200 Angstroms. In this way, the overhang of the T-shaped or Gamma-shaped gate finger **104** is separated from the semiconductor transistor structure **12** by the thickness of the first dielectric layer **32**. The first dielectric layer **32** may be a SiN layer. Other suitable dielectric materials may be used without deviating from the scope of the present disclosure. For instance, the first dielectric layer **32** may be SiO.sub.2, Si, Ge, MgOx, MgNx, ZnO, SiNx, SiOx, alloys or layer sequences thereof, or epitaxial materials.
[0077] In some aspects of the present disclosure, a second dielectric layer (not shown) may be deposited on the first dielectric layer **32**. The second dielectric layer may be the same dielectric material or a different dielectric material relative to the first dielectric layer **32**. For instance, the second dielectric layer may be a SiN layer. Other suitable dielectric materials may be used without deviating from the scope of the present disclosure. For instance, the second dielectric layer may be SiO.sub.2, Si, Ge, MgOx, MgNx, ZnO, SiNx, SiOx, alloys or layer sequences thereof, or epitaxial materials. In some embodiments, the thickness of the second dielectric layer may be about 2800

Angstroms or less, such as in a range of about 1500 Angstroms to about 2800 Angstroms, such as about 2100 Angstroms.

[0078] In some embodiments, the thickness of the first and second dielectric layers may be about 3600 Angstroms or less, such as in a range of about 3000 Angstroms to about 3600 Angstroms, such as about 3300 Angstroms.

[0079] One or more field plates (not shown) may be formed on the second dielectric layer. At least a portion of the field plate overlaps the gate finger **104**. At least a portion of the field plate is formed on a portion of the second dielectric layer. In some aspects of the present disclosure, the field plate is conductively coupled to the gate finger **104**. The field plate may reduce the peak electric field in the unit cell transistor **102** of the HEMT device **100**, which results in increased breakdown voltage and reduced charge trapping. The reduction of the electric field may also yield other benefits such as reduced leakage currents and enhanced reliability. Field plates and techniques for forming field plates are discussed, by way of example, in U.S. Pat. Nos. 7,550,783 and 8,120,064, the disclosures of which are incorporated by reference herein in their entireties. In embodiments in which a field plate or other structure is formed, the source fingers **108**, gate bond pad **110**, and drain bond pad **112** are the highest, or uppermost, structures in the semiconductor device. In some designs, the gate bond pad **110**, and drain bond pad **112** may connect to gate fingers **104** and drain fingers **106** through vias, as well known in the art, and not further elaborated herein.

[0080] A HEMT transistor cell **102** may be formed by the active region between the source finger **108** and the drain finger **106** under the control of the gate finger **104** between the source finger **108** and the drain finger **106**. FIG. **6** depicts a cross-sectional view of one unit or cell **102** of an HEMT device **100** for purposes of illustration. As shown in FIG. **1**, the HEMT device unit or cell **102** may be formed adjacent to additional HEMT device cells and may share, for instance, a source finger **108** with adjacent HEMT device cells **102**. The HEMT device **100** may be formed, at least in part, via wafer-level processing, and the wafer may then be diced to provide a plurality of individual HEMT devices **100**.

[0081] In some aspects of the present disclosure, the semiconductor transistor structure **12** may include implanted regions under the source finger **108** and drain finger **106** (not shown). The implanted regions include a distribution of implanted dopants extending into the channel layer **16**. The implanted dopants may provide, for instance, for N-type doping of the semiconductor transistor structure **12**. The implanted dopants may be, for instance, silicon or germanium dopants. The implanted region may have a distribution of implanted dopants that extends to a depth from a surface (e.g., a top surface) of the semiconductor transistor structure **12**. The depth may be about 200 Angstroms or greater, such as about 250 Angstroms or greater, such as in a range of about 200 Angstroms to about 300 Angstroms. For instance, in some aspects of the present disclosure, the implanted region **302** can extend to a depth of about 200 Angstroms to about 300 Angstroms beneath a surface of the Group III-nitride semiconductor structure. In particular, the implanted region can extend completely through the barrier layer **18** and into the channel layer **16**. The distribution of implanted dopants may extend through other layers, if present, such as a cap layer.

[0082] The implanted regions may have a peak dopant concentration. In some aspects of the present disclosure, a peak dopant concentration of the distribution of implanted dopants in the implanted regions is in the channel layer **16**. For instance, the implanted regions may be doped such that the distribution of implanted dopants is greater at a portion of the implanted regions at a depth of the channel layer **16** than at a portion of the implanted regions at a depth of the barrier layer **18** (and/or another layer). In some examples, the distribution of implanted dopants may provide a substantially uniform concentration of implanted dopants throughout the implanted regions of the semiconductor transistor structure **12**. In some aspects of the present disclosure, a peak dopant concentration of the distribution of implanted dopants can be about 1×10.sup.18 dopants/cm.sup.3.

[0083] The RF amplifier with integrated harmonic termination circuit **118** may be packaged in a variety of ways. FIG. **7** shows a Group III nitride-based RF amplifier **52**, which includes a unit cell configuration of parallel HEMTs **100** mounted within an open cavity package **54**. The package **54** includes a gate lead **56**, a drain lead **58**, a metal flange **60** and a ceramic sidewall and lid **62**. The HEMT **100** is mounted on the upper surface of the metal flange **60** in a cavity formed by the metal flange **60** and the ceramic sidewall and lid **62**.

[0084] Input matching circuits **64** and/or output matching circuits **66** may also be mounted within the housing **62**. The matching circuits **64**, **66** may be impedance matching circuits that match the impedance of the fundamental component of RF signals input to or output from the RF transistor amplifier **52** to the impedance at the input or output of the HEMT **100**, respectively. As schematically shown in FIG. **7**, the input and output matching circuits **64**, **66** may be mounted on the metal flange **60**. The gate lead **56** may be connected to the input matching circuit **64** by one or more first bond wires **68**, and the input matching circuit **64** may be connected to the gate bond pad **110** of HEMT **100** by one or more second bond wires **70**. Similarly, the drain lead **58** may be connected to the output matching circuit **66** by one or more fourth bond wires **72**, and the output matching circuit **66** may be connected to the drain bond pad **112** of HEMT **100** by one or more third bond wires **74**.

[0085] The source finger **108** of the HEMT **100** may connect to the metal layer **30** on the lower side of the substrate **14**, which may directly contact the metal flange **60**. As shown, the harmonic termination circuit **118** described herein comprises a MIMcap **122** formed on a source finger **108**, and a bond wire **120** connecting the MIMcap **122** to, in this case, the gate bond pad **110**. The metal flange **60** may provide the electrical connection to the source finger **108** and may also serve as a heat dissipation structure. The first through fourth bond wires **68**-**74** may form part of the input and/or output matching circuits. The housing **62** may comprise a ceramic housing, and the gate lead **56** and the drain lead **58** may extend through the housing **52**. The housing **52** may comprise multiple pieces, such as a frame that forms the lower portion of the sidewalls and supports the gate and drain leads **56**, **58**, and a lid that is placed on top of the frame. The interior of the device may comprise an air-filled cavity.

[0086] The metal flange **60** may act as a heat sink that dissipates heat that is generated in the HEMT **100**. The heat is primarily generated in the upper portion of the HEMT **100** where relatively high current densities are generated in, for example, the channel regions of the unit cell transistors **102**. This heat may be transferred though both the source vias **114** and the substrate **116** to the metal flange **60**.

[0087] FIG. **8** is a schematic side view of a conventional packaged Group III nitride-based RF transistor amplifier **52**′ that is similar to the RF transistor amplifier **52** discussed above with reference to FIG. **7**. RF transistor amplifier **52**′ differs from RF transistor amplifier **52** in that it includes a different package **54**′. The package **54**′ includes a metal submount **76** (which acts as a metal heat sink and can be implemented as a metal slug), as well as gate and drain leads **56**′, **58**′. In some embodiments, a metal lead frame may be formed that is then processed to provide the metal submount **76** and/or the gate and drain leads **56**′, **58**′. RF transistor amplifier **52**′ also includes a plastic overmold **62**′ that at least partially surrounds the HEMT **100**, the leads **56**′, **58**′ and the metal submount **76**. The plastic overmold **62**′ replaces the ceramic sidewalls and lid **62** included in RF transistor amplifier **52**.

[0088] Depending on the embodiment, the packaged transistor amplifier **52**′ can include, for example, a monolithic microwave integrated circuit (MMIC) as the HEMT **100** in which case the HEMT **100** incorporates multiple discrete devices. When the HEMT **100** is a MMIC implementation, the input matching circuits **64** and/or the output matching circuits **66** may be omitted (since they may instead be implemented within the HEMT **100**) and the bond wires **68** and/or **72** may extend directly from the gate and drain leads **56**′, **58**′ to the gate bond pad **50** and drain metal contact **48**. In some embodiments, the packaged RF transistor amplifier **52** can include multiple RF transistor amplifier die that are connected in series to form a multiple stage RF transistor amplifier and/or may include multiple transistor die that are disposed in multiple paths (e.g., in parallel) to form an RF transistor amplifier with multiple RF transistor amplifier die and multiple paths, such as in a Doherty amplifier configuration.

[0089] In other cases, Group III nitride-based RF amplifiers may be implemented as monolithic microwave integrated circuit ("MMIC") devices in which one or more RF amplifier die(s) are implemented together with their associated impedance matching and harmonic termination circuits in a single, integrated circuit die. Examples of such Group III nitride-based RF amplifiers are disclosed, for example, in U.S. Pat. No. 9,947,616, the entire content of which is incorporated herein by reference.

[0090] FIG. **9** is a cross-sectional view illustrating an example of a thermally enhanced integrated circuit device package, more specifically a T3PAC package. The T3PAC package **52**″ of FIG. **9** can be a ceramic-based package that includes a base **78** and an upper housing with a lid member **80** and sidewall members **82**. The lid member **80** and sidewalls **82** similarly define an open cavity surrounding the HEMT **100** on the conductive base or flange **78**, which likewise provides both an attachment surface and thermal conductivity (e.g., a heat sink) for dissipating or otherwise transmitting heat outside of the package **600**″.

[0091] The flange **78** can be an electrically conductive material, for example, a copper layer/laminate or an alloy or metal-matrix composite thereof. In some embodiments, the flange **78** may include a copper-molybdenum (CuMo) layer, CPC (Cu/MoCu/Cu), or other copper alloys, such copper-tungsten CuW, and/or other laminate/multi-layer structures. In the example of FIG. **9**, the flange **78** is illustrated as a copper-molybdenum (RCM60)-based structure to which the sidewalls **82** and/or lid member **80** are attached, e.g., by a conductive glue **1208**.

[0092] The flange **78** also provides the source lead for the package **52**″. The gate lead **56** and drain lead **58** are provided by respective conductive wiring structure **86** which is attached to the flange **78** and supported by the sidewall members **82**.

[0093] Aspects of the present invention present numerous advantages over the prior art. Component count is reduced, compared to solutions that provide harmonic termination circuitry off-chip. Where harmonic termination circuitry is integrated, the capacitor structure and inductor traces or transmission line must be formed separately from the RF amplifier, which increases the size, and hence cost, of the die. By integrating the inventive harmonic termination circuit **188** into the RF amplifier **100**, both component count and die area are reduced, both of which save cost. The MIMcaps **122** are inherently high Q (low loss), leading to improved harmonic termination and better efficiency compared to off-chip capacitors. The bond wires are inherently high Q (low loss), leading to improved harmonic termination and better efficiency compared to on-chip spiral inductors or transmission lines. The harmonic termination circuit **188** is easily tuned to modify the harmonic termination frequency, by adjusting the length of the bond wires. This eliminates the need to re-design the circuit and re-fabricate the die in case the harmonic design was slightly off, and allows for broader reuse of the die, such as for applications operating a different fundamental frequency. The technique can be applied to the input, the output, or both. Multiple harmonics can be terminated at either or both of the input and output.

[0094] The present invention may, of course, be carried out in other ways than those specifically set forth herein without departing from essential characteristics of the invention. The present embodiments are to be considered in all respects as illustrative and not restrictive, and all changes coming within the meaning and equivalency range of the appended claims are intended to be embraced therein. Although steps of various processes or methods described herein may be shown and described as being in a sequence or temporal order, the steps of any such processes or methods are not limited to being carried out in any particular sequence or order, absent an indication otherwise. Indeed, the steps in such processes or methods generally may be carried out in various different sequences and orders while still falling within the scope of the present invention.

Claims

1. A semiconductor transistor configured to amplify a Radio Frequency (RF) signal at a fundamental frequency, comprising: a substrate having upper and lower surfaces; a unit cell semiconductor transistor structure formed on the upper surface of the substrate; source and drain fingers formed directly on, and electrically connected to, the semiconductor transistor structure; gate fingers formed over the semiconductor transistor structure in active areas between the source and drain fingers; gate and drain bond pads connected to respective gate and drain fingers; wherein the source fingers are electrically connected to RF signal ground; and a series resonant circuit connected between one of the gate and drain bond pads and a source finger, the series resonant circuit comprising a Metal-Insulator-Metal (MIM) capacitor formed on the source finger; and an inductance implemented as a bond wire connected between the gate or drain bond pad and the MIM capacitor.
2. The transistor of claim 1 wherein the series resonant circuit is configured to present a low impedance path to RF signal ground at a selected harmonic of the fundamental frequency.
3. The transistor of claim 2 wherein the series resonant circuit is configured to be tuned by adjusting a length of the bond wire.
4. The transistor of claim 3 wherein the length of the bond wire is adjusted by varying its height.
5. The transistor of claim 3 wherein the length of the bond wire is adjusted after fabrication of the semiconductor transistor is otherwise complete.
6. The transistor of claim 1 further comprising: a via through the substrate and electrically connected to a conductive layer formed on the lower surface of the substrate; wherein the conductive layer is electrically connected to RF signal ground; and wherein the source terminal is formed over, and electrically connected to, the via.
7. The transistor of claim 1 wherein the MIM capacitor comprises: the source terminal; an insulator deposited over at least part of the source terminal; and a metal layer deposited over the insulator.
8. The transistor of claim 7 wherein the bond wire is connected to the metal layer.
9. A method of manufacturing a semiconductor transistor configured to amplify a Radio Frequency (RF) signal at a fundamental frequency, comprising: providing a substrate having upper and lower surfaces; forming a unit cell semiconductor transistor structure on the upper surface of the substrate; forming source and drain fingers directly on, and electrically connected to, the semiconductor transistor structure; forming gate fingers over the semiconductor transistor structure in active areas between the source and drain fingers; forming gate and drain bond pads connected to respective gate and drain fingers; electrically connecting the source fingers to RF signal ground; and forming a series resonant circuit connected between one of the gate and drain bond pads and a source finger, comprising forming a Metal-Insulator-Metal (MIM) capacitor on the source finger; and connecting a bond wire between the gate or drain bond pad and the MIM capacitor.
10. The method of claim 9 wherein the series resonant circuit is configured to present a low impedance path to RF signal ground at a selected harmonic of the fundamental frequency.
11. The method of claim 10 further comprising tuning the series resonant circuit by adjusting a length of the bond wire.
12. The method of claim 11 wherein adjusting a length of the bond wire comprises varying its height.
13. The method of claim 11 wherein tuning the series resonant circuit occurs after fabrication of the semiconductor transistor is otherwise complete.
14. The method of claim 9 further comprising: a via through the substrate and electrically connected to a conductive layer formed on the lower surface of the substrate; wherein the conductive layer is electrically connected to RF signal ground; and wherein the source terminal is formed over, and electrically connected to, the via.

15. The method of claim 9 wherein the MIM capacitor comprises: the source terminal; an insulator deposited over at least part of the source terminal; and a metal layer deposited over the insulator.
16. The method of claim 15 wherein the bond wire is connected to the metal layer.